# HRIDEY MARWAH

Delhi, India |

## OBJECTIVE

Computer Science (AI) student researching and building AI systems, multi-agent architectures, and full-stack applications. Exploring reliable AI orchestration, agentic coding systems, reasoning, memory, tool use, and verification while contributing to open-source machine-learning infrastructure.

## EDUCATION

**Newton School of Technology** **2025–2029**
B.Tech in Computer Science & Engineering (AI)
Rishihood University, Sonipat

## SKILLS

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|---|---|
| **Languages** | Python, JavaScript, TypeScript, SQL, HTML, CSS |
| **Frontend** | React.js, Next.js, Tailwind CSS |
| **Backend** | Node.js, Express.js, REST APIs |
| **Databases** | PostgreSQL, Prisma |
| **AI / ML** | JAX, NumPy, OCR, Computer Vision, LLMs, Multi-Agent Systems |
| **Tools** | Git, GitHub, VS Code, Docker, Ollama |

## PROJECTS

### Brain Architecture — Research on Cognitive AI Orchestration

Researching and prototyping a brain-inspired multi-agent architecture for reliable multi-step task execution, exploring planning, specialized agents, memory, reasoning, tool use, critique, synthesis, execution, and verification through iterative feedback loops.

### Co-Team + Forge — Multi-Agent Coding Harness

Built a multi-agent coding architecture separating high-level reasoning from controlled code execution. Designed orchestration across planning, research, critique, synthesis, execution, and verification, with a dedicated coding harness for tool interaction and feedback-driven development workflows.

### Supply Chain Disruption Simulator

*React, Node.js, PostgreSQL, Prisma*

Built an interactive supply-chain simulation platform modeling disruption propagation across directed dependency networks. Implemented graph-based modeling and simulated failures, capacity reductions, route blockages, shipment delays, inventory effects, and recovery strategies.

### Aushadhi OCR — Medical Prescription Digitization

*Python, OpenCV, OCR, Computer Vision*

Built an OCR pipeline for extracting structured information from medical prescriptions. Applied image preprocessing and AI-based correction to improve extraction reliability from noisy prescription images.

## EXPERIENCE

**Open Source Contributor — JAX** **2026–Present**

- Contributing to JAX, a high-performance numerical computing framework.
- Working with large-scale open-source codebases and performance-oriented systems.
- Exploring optimization techniques relevant to machine-learning workloads.

## ACHIEVEMENTS

- Hackathon Finalist — IIIT Gwalior, InfoTsav'25
- Hackathon Finalist — IISc Bangalore Hackathon
- Hackathon Finalist — Decode by NST
- GSoC 2026 Applicant — Open Source Program